Course: BM20A4703 Partial Differential Equations with Applications.
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Worksheet 6

Exercise 6.1

(i) [10] Let $u = u(x_1, x_2)$ be a smooth function of compact support in $\mathbb{R}^2$. Prove that

$$\max_{x_i} |u^m| \leq m \int_{-\infty}^{\infty} |u^{m-1} u_{x_i}| \mathrm{d}x_i, \quad (i = 1, 2),$$

for any $m \in \mathbb{N}, m \geq 1$.

(ii) [20] Prove that for any smooth function u with compact support in some domain $\Omega \subset \mathbb{R}^2$, we have

$$\|u\|^4_{L^4(\Omega)} \leq 2\|u\|^2_{L^2(\Omega)} \|\nabla u\|^2_{L^2(\Omega)}.$$

Hint: Use the fact that $u^2 u^2 \leq \max_{x_1} u^2 \max_{x_2} u^2$ and integrate both sides over $\mathbb{R}^2$, then use (i) and Cauchy-Schwartz inequality.

Exercise 6.2

[20] Let $E(t)$ be the kinetic energy

$$E(t) := \int_\Omega \left(\frac{1}{2} \rho |u|^2 \right) \mathrm{d}x = \frac{1}{2} \rho \|u\|^2_{L^2(\Omega)}, \tag{1}$$

for the Navier–Stokes equations ($\nu > 0$)

$$\frac{\partial u}{\partial t} + u \cdot \nabla u + \frac{1}{\rho} \nabla p = \nu \Delta u, \tag{2}$$

$$\nabla \cdot u = 0, \tag{3}$$

with zero the boundary condition $n \cdot u|_{\partial\Omega} = 0$. Prove that the kinetic energy is dissipative, that is,

$$\frac{\mathrm{d}}{\mathrm{d}t} E(t) \leq 0.$$

Hint: Use integration by parts.

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